Circulation: Heart Failure

ORIGINAL ARTICLE

Finerenone, Liver Biomarkers, and Heart Failure With Mildly Reduced/Preserved Ejection Fraction: An Analysis of FINEARTS-HF

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BACKGROUND: The prevalence and prognostic significance of liver biomarkers in heart failure (HF) with mildly reduced or preserved ejection fraction are uncertain, with both potential hemodynamic and metabolic contributions to liver dysfunction in these patients. We evaluated the prevalence and prognostic value of liver biomarkers and assessed the effects of the nonsteroidal mineralocorticoid receptor antagonist finerenone on these biomarkers and clinical outcomes in FINEARTS-HF (Finerenone Trial to Investigate Efficacy and Safety Superior to Placebo in Patients With Heart Failure).

METHODS: FINEARTS-HF was a randomized, double-blind, placebo-controlled trial that enrolled 6001 patients with left ventricular ejection fraction ≥40%, evidence of structural heart disease, and elevated NT-proBNP (N-terminal pro-B-type natriuretic peptide) levels. Liver biomarkers examined were total bilirubin, alkaline phosphatase, alanine aminotransferase, and aspartate aminotransferase.

RESULTS: Among 5873 patients with available baseline bilirubin measurements, 11.9% had elevated levels (>1.0 mg/dL). Higher bilirubin levels were associated with a greater risk of total worsening HF events and cardiovascular death. Compared with placebo, finerenone rapidly reduced bilirubin and alkaline phosphatase levels (but not transaminase levels), with effects sustained over time. Finerenone reduced the risk of total worsening HF events and cardiovascular death across all bilirubin tertiles (T1 [<0.4 mg/dL], rate ratio 0.94 [95% CI, 0.75–1.17]; T2 [0.5–0.6 mg/dL], 0.83 [0.66–1.05]; T3 [≥0.7 mg/dL], 0.77 [0.62–0.97]), with no significant interaction by bilirubin level ($P_{interaction}$=0.43). Consistent effects were observed for the components of the primary outcome, all-cause death, and improvement in the Kansas City Cardiomyopathy Questionnaire total symptom score.

CONCLUSIONS: Baseline bilirubin concentration was an independent predictor of worse outcomes but did not modify the benefits of finerenone on morbidity and mortality in HF with mildly reduced or preserved ejection fraction. Finerenone reduced bilirubin and alkaline phosphatase, suggesting a possible decongestive effect in HF with mildly reduced or preserved ejection fraction.

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Key Words: bilirubin ■ finerenone ■ heart failure ■ liver function tests

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WHAT IS NEW?

- In a secondary analysis of FINEARTS-HF (Finerenone Trial to Investigate Efficacy and Safety Superior to Placebo in Patients With Heart Failure), baseline bilirubin level was an independent predictor of worse outcomes.
- Compared with placebo, the nonsteroidal mineralocorticoid receptor antagonist, finerenone, reduced the risk of worsening heart failure events and cardiovascular deaths and improved symptoms across the range of bilirubin concentrations measured at baseline.
- Finerenone led to a rapid and sustained reduction in levels of bilirubin and ALP (alkaline phosphatase), but not transaminases.

WHAT ARE THE CLINICAL IMPLICATIONS?

- These findings suggest that finerenone is an effective and well-tolerated treatment in patients with heart failure with mildly reduced or preserved ejection fraction, regardless of bilirubin level.

Nonstandard Abbreviations and Acronyms

ALP	alkaline phosphatase
ALT	alanine aminotransferase
AST	aspartate aminotransferase
DAPA-HF	Dapagliflozin and Prevention of Adverse Outcomes in Heart Failure
eGFR	estimated glomerular filtration rate
EMPEROR-Preserved	Empagliflozin Outcome Trial in Patients With Chronic Heart Failure With Preserved Ejection Fraction
FINEARTS-HF	Finerenone Trial to Investigate Efficacy and Safety Superior to Placebo in Patients With Heart Failure
HF	heart failure
HFmrEF	heart failure with mildly reduced ejection fraction
HFpEF	heart failure with preserved ejection fraction
HFrEF	heart failure with reduced ejection fraction
KCCQ-TSS	Kansas City Cardiomyopathy Questionnaire total symptom score
LVEF	left ventricular ejection fraction
NT-proBNP	N-terminal pro-B-type natriuretic peptide
NYHA	New York Heart Association
PARADIGM-HF	Prospective Comparison of ARNI With ACEI to Determine Impact on Global Mortality and Morbidity in Heart Failure
SGLT2	sodium-glucose cotransporter 2

Liver biomarkers such as bilirubin and hepatic transaminases are frequently abnormal in patients with heart failure (HF) and reduced ejection fraction (HFrEF), especially during episodes of decompensation.[1–8] Increases in these biomarkers are most often attributed to liver congestion and hypoperfusion, respectively.[9–11] Elevation of bilirubin, but not transaminases, is consistently associated with worse clinical outcomes in HFrEF.[1–8] More recently, this has also been demonstrated in ambulatory HFrEF cohorts, even when bilirubin is added to multivariable models incorporating NT-proBNP (N-terminal pro-B-type natriuretic peptide).[1,2] Much less is known about the prevalence and prognostic significance of abnormal liver biomarkers in patients with HF and HF with preserved ejection fraction (HFpEF).[12,13] In addition to hemodynamic determinants, patients with HFpEF may have abnormalities of liver biomarkers related to the accumulation of liver fat associated with type 2 diabetes, obesity, and related nonalcoholic fatty liver disease.[14,15] Both the hemodynamic and potential metabolic drivers of disturbances in liver biomarkers in HFpEF may, in the long term, also lead to hepatic inflammation and fibrosis, potentially involving aldosterone and the mineralocorticoid receptor.[16–21] Indeed, preclinical data have suggested that activation of the mineralocorticoid receptor, which is expressed not only in the kidney and the heart, but also in hepatic tissue (including hepatocytes and hepatic stellate cells), may promote hepatic steatosis, inflammation, and fibrosis, particularly in the setting of metabolic dysfunction.[16–21] Conversely, antagonizing the mineralocorticoid receptor has been shown to attenuate these pathological processes by reducing oxidative stress, hepatic fibrosis, and insulin resistance.[16–21] Because of these effects and their decongestive properties, mineralocorticoid receptor antagonists are of particular interest in patients with HFpEF and abnormal liver biomarkers.[22–24]

Therefore, we investigated the prevalence of elevated liver biomarkers in the FINEARTS-HF trial (Finerenone Trial to Investigate Efficacy and Safety Superior to Placebo in Patients With Heart Failure), which demonstrated that the nonsteroidal mineralocorticoid receptor antagonist, finerenone, compared with placebo, reduced the risk of the primary composite outcome of total (first and

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repeat) worsening HF events and cardiovascular death in 6001 patients with a left ventricular ejection fraction (LVEF) ≥40%.[25] We hypothesized that bilirubin would be predictive of outcomes and that finerenone, compared with placebo, would lead to improvements in both bilirubin and transaminase concentrations.

METHODS

FINEARTS-HF was a randomized, double-blind, placebo-controlled trial in patients with symptomatic HF with mildly reduced ejection fraction (HFmrEF) or HFpEF, investigating the efficacy and safety of finerenone compared with a matching placebo in addition to usual therapy. The design, baseline characteristics, and primary results of FINEARTS-HF are published.[25–27] The trial protocol was approved by the Ethics Committee at all participating institutions, and all patients provided written informed consent. The corresponding author had full access to all the trial data and takes responsibility for its integrity and the data analysis. Trial data will be made available by the sponsor, Bayer, in accordance with their data sharing policy.

Trial Patients

Key inclusion criteria were age ≥40 years, diuretic treatment for ≥30 days before randomization, New York Heart Association (NYHA) functional class II to IV, LVEF ≥40%, evidence of structural heart disease (either left atrial enlargement or left ventricular hypertrophy), and elevated natriuretic peptide levels. Both ambulatory and hospitalized patients were eligible for enrollment. Patients with prior LVEF <40% with subsequent improvement to ≥40% were also eligible for enrollment, provided that ongoing HF symptoms were present. Key exclusion criteria were an estimated glomerular filtration rate (eGFR)<25 mL/min per 1.73 m^2, serum/plasma potassium >5.0 mmol/L, or hepatic impairment, classified as Child-Pugh class C, at screening or randomization. A complete list of exclusion criteria is provided in the design paper.[27]

Eligible participants were randomized in a 1:1 ratio to finerenone or a matching placebo. Participants with an eGFR ≤60 mL/min per 1.73 m^2 started 10 mg once daily with a maximum maintenance dose of 20 mg once daily, whereas participants with an eGFR >60 mL/min per 1.73 m^2 started 20 mg once daily with a maximum maintenance dose of 40 mg once daily.

Trial Outcomes

The primary outcome in FINEARTS-HF was the composite of cardiovascular death and total (first and recurrent) HF events (ie, HF hospitalization or urgent HF visit). The secondary outcomes were total (first and recurrent) HF events; improvement in NYHA class from baseline to 12 months; change in the Kansas City Cardiomyopathy Questionnaire (KCCQ) total symptom score (KCCQ-TSS) from baseline to 6, 9, and 12 months; composite kidney end point (defined as sustained decrease in eGFR ≥50% relative to baseline over at least 4 weeks, or sustained eGFR decline <15 mL/min per 1.73 m^2, or initiation of dialysis or renal transplantation); and all-cause death. All deaths and potential primary nonfatal events were adjudicated by an independent clinical events committee. The composite kidney outcome was not explored further in the present analysis because there were few events overall, making subgroup analysis unreliable.

Prespecified safety analyses included hyperkalemia, hypokalemia, hypotension, and elevations in serum creatinine levels. Safety analyses were only performed in patients who had received at least 1 dose of either finerenone or placebo.

Liver Biomarkers

Total bilirubin, ALP (alkaline phosphatase), ALT (alanine aminotransferase), and AST (aspartate aminotransferase) were measured at baseline, 1 month, 3, months, 6 months, 9 months, and 12 months, and every fourth month hereafter. Samples were processed in a central laboratory.

Cutoff values to define elevated total bilirubin, ALP, ALT, and AST levels were >1.0 mg/dL, >120 U/L, >35 U/L, and >35 U/L, respectively.[28]

Bilirubin was the focus of our analysis, given the evidence that it is the most prognostically important liver test. Patients were divided into 3 subgroups based on the tertiles of baseline bilirubin. In the analyses of the association between bilirubin as a categorical variable and outcomes, the middle tertile was used as the reference, given the suggestion of a J-shaped relationship between bilirubin level and outcomes in patients with HFmrEF/HFpEF enrolled in the EMPEROR-Preserved trial (Empagliflozin Outcome Trial in Patients With Chronic Heart Failure With Preserved Ejection Fraction).[12]

Statistical Analyses

Baseline characteristics according to tertiles of baseline bilirubin were summarized as frequencies with percentages, means with standard deviations, or medians with interquartile ranges, and differences were tested using the χ^2 test for binary or categorical variables and the Kruskal-Wallis test and ANOVA test for non-normal and normally distributed continuous variables, respectively.

The association between tertiles of baseline bilirubin and clinical outcomes was evaluated using Cox proportional-hazards models for time-to-event data and semiparametric proportional-rates models for total (first and recurrent) events,[29] and after the prespecified statistical analysis plan, hazard ratios and rate ratios, respectively, were stratified according to geographic region and LVEF stratification (<60%, ≥60%) and adjusted for treatment assignment. In addition, hazard ratios and rate ratios, stratified by geographic region and LVEF stratification and adjusted for treatment assignment, age, sex, systolic blood pressure, heart rate, body mass index, eGFR, LVEF, NYHA functional class, log of NT-proBNP, log of high-sensitivity troponin T, prior HF hospitalization, type 2 diabetes, myocardial infarction or coronary revascularization, and atrial fibrillation/flutter, were reported. The relationship between liver biomarkers as continuous variables and the risk of outcomes was also examined in restricted cubic spline analyses with the median value as reference (unless otherwise stated). In addition, in a landmark analysis, the association between absolute changes in liver biomarkers from baseline to visit 3 (ie, ≈3 months after randomization) as continuous variables and clinical outcomes was examined in restricted cubic spline analyses with no change as reference.

To compare the effects of finerenone versus placebo on clinical outcomes according to tertiles of baseline bilirubin,

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time-to-event data and total events were evaluated with Cox proportional-hazards models and semiparametric proportional-rates models, respectively, and these models were stratified according to geographic region and LVEF stratification. The effect of finerenone was also examined according to continuous liver biomarker levels using fractional polynomial models, restricted cubic spline models, or linear regression models, whichever had the lowest Akaike Information Criterion score.

The proportion of patients with improvement in NYHA class from baseline to 12 months according to tertiles of baseline bilirubin was analyzed using a logistic regression model, adjusted for geographic region and LVEF stratification, and odds ratios were reported.

The change in KCCQ-TSS from baseline to 12 months was summarized as means and standard deviations within each tertile of baseline bilirubin at 12 months, and the effect of finerenone versus placebo on the change in KCCQ-TSS from baseline to 12 months was estimated using a linear regression model within each tertile of baseline bilirubin, adjusted for baseline KCCQ-TSS, geographic region, and LVEF stratification.

The change in liver biomarkers from baseline to 1 month and 12 months after randomization, respectively, was analyzed using ANCOVA models, adjusted for baseline value, geographic region, and LVEF stratification, and results were reported as the ratio of geometric means with 95% CIs.

Region and LVEF category were included as covariates (ie, adjusted for) in the logistic and linear regression models to account for the stratified randomization in FINEARTS-HF.

The Wald test was used to test for interaction between the treatment effect of finerenone and bilirubin for all efficacy end points, and the respective models included treatment assignment, tertiles of bilirubin (included as a categorical variable), and their interaction as covariates, in addition to those described above. For the safety outcomes, the Wald test was used to test for interaction between the treatment effect of finerenone and bilirubin in a logistic regression model, which included treatment assignment, tertiles of bilirubin (included as a categorical variable), and their interaction as covariates.

All analyses were conducted using STATA version 18.0 (College Station, TX).

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RESULTS

A measurement of total bilirubin, ALP, ALT, and AST at baseline was available in 5873 (97.9%), 5856 (97.6%), 5814 (96.9%), and 5725 (95.4%) participants, respectively. The median values were 0.5 mg/dL (0.4–0.8), 79 U/L (65–98), 20 U/L (17–26), and 17 U/L (13–24), respectively. Total bilirubin, ALP, ALT, and AST levels were elevated in 11.9%, 10.8%, 8.6%, and 7.8% of patients, respectively.

Baseline Characteristics According to Tertiles of Bilirubin

The baseline characteristics of patients according to tertiles of bilirubin are shown in Table 1. Compared with patients with lower levels, those with higher bilirubin levels were older, more often male, and less often White, and they had lower systolic blood pressure, proportional pulse pressure, LVEF, blood urea nitrogen, and urine albumin-to-creatinine ratio, but a higher heart rate, eGFR, hemoglobin, and NT-proBNP. They also had elevated levels of other liver biomarkers.

Compared with patients with lower levels, those with higher bilirubin levels were more likely to have a prior HF hospitalization, and atrial fibrillation/flutter, but they were less likely to be obese and to have hypertension, type 2 diabetes, and sleep apnea. Although there was no significant difference in NYHA functional class across bilirubin tertiles, patients with higher bilirubin generally had higher (better) KCCQ scores compared with those with lower bilirubin.

Regarding pharmacological therapy, patients with higher bilirubin were more frequently treated with ARNI, a β-blocker, an anticoagulant, and digoxin and less frequently with an SGLT2 (sodium-glucose cotransporter 2) inhibitor.

Clinical Outcomes According to Liver Biomarkers

Total Bilirubin

In the minimally adjusted models, compared with patients in bilirubin tertile 2, those in tertile 3 had a significantly higher risk of all outcomes except for cardiovascular and all-cause death. There was no difference in outcomes between tertiles 1 and 2 (Table 2). In the fully adjusted models, compared with patients in bilirubin tertile 2, those in bilirubin tertile 3 no longer had a significantly higher risk of any of the outcomes examined.

When examined as a continuous variable, bilirubin levels higher than the median value were associated with a greater risk of the primary outcome, each of its components, but not all-cause death, after adjustment for prognostic variables (Figure 1). Bilirubin levels lower than the median value were associated with a higher risk of all-cause death.

Other Liver Biomarkers

After adjustment for prognostic variables, ALP levels higher than the median value were associated with a greater risk of the primary outcome, each of its components, and all-cause death (Figure S1). Although neither ALT nor AST were associated with the primary outcome or total worsening HF events, levels lower than the median value of both markers were associated with a higher risk of all-cause death (Figure S1).

Effects of Finerenone on Clinical Outcomes According to Liver Biomarkers

Total Bilirubin

Compared with placebo, finerenone reduced the risk of total (first and recurrent) worsening HF events and

Table 1. Baseline Characteristics According to Tertiles of Bilirubin

	Tertile 1; ≤0.4 mg/dL (n=1923)	Tertile 2; 0.5–0.6 mg/dL (n=1932)	Tertile 3; ≥0.7 mg/dL (n=2018)	*P* value
Age, y, mean (SD)	71.4±9.9	72.1±9.7	72.2±9.4	0.023
Sex, n (%)				<0.001
Men	817 (42.5)	1086 (56.2)	1302 (64.5)	
Women	1106 (57.5)	846 (43.8)	716 (35.5)	
Race, n (%)				<0.001
White	1568 (81.5)	1532 (79.3)	1515 (75.1)	
Black	56 (2.9)	19 (1.0)	13 (0.6)	
Asian	233 (12.1)	334 (17.3)	424 (21.0)	
Other	66 (3.4)	47 (2.4)	66 (3.3)	
Geographic region, n (%)				<0.001
Western Europe, Oceania, and Others	444 (23.1)	408 (21.1)	351 (17.4)	
Eastern Europe	756 (39.3)	902 (46.7)	944 (46.8)	
Asia	228 (11.9)	334 (17.3)	416 (20.6)	
North America	197 (10.2)	129 (6.7)	130 (6.4)	
Latin America	298 (15.5)	159 (8.2)	177 (8.8)	
Physiological measures				
Systolic blood pressure, mm Hg, mean (SD)	130.8±15.5	129.5±15.0	128.0±15.5	<0.001
Diastolic blood pressure, mm Hg, mean (SD)	74.2±10.2	75.5±10.2	76.6±10.5	<0.001
Proportional pulse pressure, mean (SD)	0.43±0.08	0.41±0.08	0.40±0.08	<0.001
Heart rate, bpm, mean (SD)	70.1±11.1	71.0±11.9	73.1±12.1	<0.001
Body mass index, mean (SD)	29.8 (26.0–34.6)	29.2 (25.5–33.5)	28.7 (25.2–32.7)	<0.001
Body mass index, n (%)				<0.001
<18.5	20 (1.0)	20 (1.0)	25 (1.2)	
18.5–24.9	372 (19.4)	396 (20.6)	444 (22.1)	
25.0–29.9	583 (30.4)	647 (33.6)	710 (35.3)	
30–34.9	496 (25.8)	519 (26.9)	501 (24.9)	
≥35.0	449 (23.4)	345 (17.9)	333 (16.5)	
Biomarkers				
NT-proBNP, pg/mL, median (IQR)	749 (359–1635)	1022 (464–1870)	1309 (603–2349)	<0.001
Atrial fibrillation/flutter on ECG	1729 (1111–2812)	1645 (1146–2739)	1756 (1170–2807)	0.42
No atrial fibrillation/flutter on ECG	548 (300–1174)	608 (318–1206)	624 (328–1405)	0.005
High-sensitivity troponin T, ng/L, median (IQR)	17.1 (6.5–28.0)	17.4 (6.5–27.9)	18.6 (6.5–28.5)	0.029
Hemoglobin A1c, %, mean (SD)	6.6±1.3	6.4±1.1	6.3±1.1	<0.001
Creatinine, μmol/L, mean (SD)	101.1±39.2	98.6±29.9	99.5±27.6	0.051
eGFR, mL/min per 1.73 m^2, mean (SD)	60.7±20.8	62.8±19.8	62.9±18.6	<0.001
eGFR, mL/min per 1.73 m^2, n (%)				<0.001
≥60	930 (48.4)	1025 (53.1)	1096 (54.3)	
<60	993 (51.6)	907 (46.9)	922 (45.7)	
UACR, mg/g, median (IQR)	18 (7–77)	17 (6–62)	19 (8–65)	0.03
UACR, mg/g, n (%)				<0.001
<30	1112 (59.7)	1180 (62.8)	1168 (59.3)	
30–299	503 (27.0)	531 (28.3)	649 (33.0)	
≥300	248 (13.3)	167 (8.9)	152 (7.7)	
Potassium, mmol/L, mean (SD)	4.4±0.5	4.4±0.5	4.3±0.5	<0.001
Sodium, mmol/L, mean (SD)	140.5±3.0	140.8±2.9	140.8±3.1	0.003
Hemoglobin, g/L, mean (SD)	127.9±15.3	134.5±15.5	138.7±16.8	<0.001
Bilirubin (total), mg/dL, mean (SD)	0.3±0.1	0.5±0.0	1.0±0.4	N/A

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Table 1. Continued

	Tertile 1; ≤0.4 mg/dL (n=1923)	Tertile 2; 0.5–0.6 mg/dL (n=1932)	Tertile 3; ≥0.7 mg/dL (n=2018)	*P* value
Bilirubin (direct), mg/dL, mean (SD)	0.1±0.0	0.2±0.1	0.3±0.1	N/A
Bilirubin (indirect), mg/dL, mean (SD)	0.2±0.1	0.4±0.1	0.7±0.3	N/A
Alkaline phosphatase, U/L, mean (SD)	85.7±34.3	83.5±30.5	88.8±36.5	<0.001
Alanine aminotransferase, U/L, mean (SD)	18.8±12.0	20.6±12.6	22.4±15.8	<0.001
Aspartate aminotransferase, U/L, mean (SD)	20.8±9.4	22.8±10.3	25.0±15.2	<0.001
Blood urea nitrogen, mg/dL, mean (SD)	23.9±10.4	22.4±9.5	21.8±8.9	<0.001
Platelet count (10^9/L), mean (SD)	241.5±74.0	217.7±62.1	201.7±63.3	<0.001
White blood cell count (10^9/L), mean (SD)	7.2±6.5	6.8±2.2	6.7±2.6	<0.001
Atrial fibrillation/flutter on ECG	420 (21.9)	756 (39.3)	1087 (54.0)	<0.001
Smoking status, n (%)				<0.001
Never	1162 (60.4)	1174 (60.8)	1273 (63.1)	
Former	561 (29.2)	593 (30.7)	611 (30.3)	
Current	200 (10.4)	165 (8.5)	134 (6.6)	
Alcohol consumption, n (%)				<0.001
Abstinent	1413 (73.5)	1346 (69.7)	1322 (65.5)	
Light	453 (23.6)	513 (26.6)	621 (30.8)	
Moderate/heavy	57 (3.0)	73 (3.8)	75 (3.7)	
LVEF, %, mean (SD)	53.7±8.0	52.5±7.7	51.6±7.7	<0.001
LVEF, %, n (%)				<0.001
<50	592 (30.8)	704 (36.5)	834 (41.3)	
50–59	887 (46.2)	866 (44.9)	852 (42.2)	
≥60	441 (23.0)	358 (18.6)	332 (16.5)	
NYHA class, n (%)				0.59
II	1334 (69.4)	1356 (70.2)	1372 (68.0)	
III	574 (29.9)	565 (29.2)	630 (31.2)	
IV	14 (0.7)	11 (0.6)	16 (0.8)	
KCCQ-TSS, mean (SD)	65.1±24.2	68.1±24.0	67.9±23.6	<0.001
KCCQ-CSS, mean (SD)	63.0±22.8	66.3±22.7	66.8±21.9	<0.001
KCCQ-OSS, mean (SD)	60.6±22.6	63.8±22.3	63.9±21.7	<0.001
Medical history, n (%)				
Hospitalization for HF	1125 (58.5)	1156 (59.8)	1258 (62.3)	0.044
Time from last HF hospitalization, n (%)				<0.001
No prior HF hospitalization	798 (41.5)	776 (40.2)	760 (37.7)	
0–7 d	262 (13.6)	312 (16.1)	405 (20.1)	
8 d–3 mo	518 (26.9)	524 (27.1)	547 (27.1)	
3–12 mo	136 (7.1)	119 (6.2)	133 (6.6)	
>1 y	209 (10.9)	201 (10.4)	173 (8.6)	
Atrial fibrillation/flutter	791 (41.1)	1074 (55.6)	1370 (67.9)	<0.001
Stroke	260 (13.5)	258 (13.4)	297 (14.7)	0.40
Myocardial infarction	505 (26.3)	518 (26.8)	495 (24.5)	0.23
PCI or CABG	656 (34.1)	663 (34.3)	689 (34.1)	0.99
Peripheral arterial occlusive disease	186 (9.7)	162 (8.4)	182 (9.0)	0.38
Hypertension	1731 (90.0)	1726 (89.3)	1757 (87.1)	0.009
Type 2 diabetes	941 (48.9)	762 (39.4)	688 (34.1)	<0.001
Chronic obstructive pulmonary disease	255 (13.3)	265 (13.7)	235 (11.6)	0.12
Sleep apnea	158 (8.2)	114 (5.9)	123 (6.1)	0.006
Liver cirrhosis	6 (0.3)	5 (0.3)	16 (0.8)	0.023

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Table 1. Continued

	Tertile 1; ≤0.4 mg/dL (n=1923)	Tertile 2; 0.5–0.6 mg/dL (n=1932)	Tertile 3; ≥0.7 mg/dL (n=2018)	*P* value
Treatment, n (%)				
ACE inhibitor	647 (33.6)	713 (36.9)	755 (37.4)	0.029
ARB	731 (38.0)	663 (34.3)	663 (32.9)	0.002
ARNI	135 (7.0)	168 (8.7)	196 (9.7)	0.009
β-blocker	1607 (83.6)	1630 (84.4)	1752 (86.8)	0.012
SGLT2 inhibitor	293 (15.2)	242 (12.5)	266 (13.2)	0.038
Loop diuretic	1669 (86.8)	1677 (86.8)	1785 (88.5)	0.19
Any diuretic	1902 (98.9)	1913 (99.0)	1990 (98.6)	0.47
Digoxin	99 (5.1)	157 (8.1)	204 (10.1)	<0.001
Statin	1323 (68.8)	1300 (67.3)	1345 (66.7)	0.34
Lipid-lowering medication	1364 (70.9)	1323 (68.5)	1367 (67.7)	0.082
Antiplatelet	870 (45.2)	739 (38.3)	671 (33.3)	<0.001
Anticoagulant	752 (39.1)	1018 (52.7)	1250 (61.9)	<0.001

ACE indicates angiotensin-converting enzyme; ARB, angiotensin receptor blocker; ARNI, angiotensin receptor–neprilysin inhibitor; CABG, coronary artery bypass graft surgery; CSS, clinical summary score; ECG, electrocardiography; eGFR, estimated glomerular filtration rate; HF, heart failure; IQR, interquartile range; KCCQ, Kansas City Cardiomyopathy Questionnaire; LVEF, left ventricular ejection fraction; N/A, not applicable; NT-proBNP, N-terminal pro-B-type natriuretic peptide; NYHA, New York Heart Association; OSS, overall summary score; PCI, percutaneous coronary intervention; SGLT2, sodium-glucose cotransporter 2; TSS, total symptom score; and UACR, urine albumin-to-creatinine ratio.

cardiovascular death across bilirubin tertiles (tertile 1 [lowest], rate ratio, 0.94 [95% CI, 0.75–1.17]; tertile 2, 0.83 [95% CI, 0.66–1.05]; tertile 3 [highest], 0.77 [95% CI, 0.62–0.97]) and, although the relative risk reduction appeared greatest in individuals with higher bilirubin, the interaction between bilirubin tertiles and the effect of treatment was not significant ($P_{interaction}$=0.43; Table 3). The effects of finerenone on secondary clinical outcomes did not differ significantly across bilirubin tertiles (Table 3).

When bilirubin was examined as a continuous variable, the beneficial effect of finerenone on the primary composite outcome and total worsening HF events appeared to be greater in those with higher bilirubin ($P_{interaction}$=0.089 and $P_{interaction}$=0.087, respectively; Figure 2). The effect of finerenone on cardiovascular or all-cause death was not modified by continuous bilirubin.

The mean increase in KCCQ-TSS from baseline to 12 months was greater with finerenone compared with placebo, and, although the improvement in KCCQ-TSS appeared greatest in patients in the highest tertile of bilirubin (tertile 1 [lowest], mean difference 1.13 [95% CI, −0.62 to 2.89]; tertile 2, 0.69 [95% CI, −0.92 to 2.30]; tertile 3 [highest], 3.15 [95% CI, 1.52–4.77]), the interaction between bilirubin and the effect of finerenone on the improvement in KCCQ-TSS from baseline to 12 months was not significant ($P_{interaction}$=0.11; Table 3). The effect of finerenone on improvement in NYHA class from baseline to 12 months was not modified by bilirubin ($P_{interaction}$=0.70).

Patients with higher bilirubin were at lower risk of increases in creatinine levels and hyperkalemia, but at higher risk of hypokalemia than those with lower bilirubin. Patients randomized to finerenone were more likely to experience increases in potassium and creatinine levels, and a decrease in systolic blood pressure (to <100 mmHg) compared with participants randomized to placebo. Conversely, finerenone reduced the risk of hypokalemia compared with placebo. All these findings were generally consistent across bilirubin categories (Table 4).

Other Liver Biomarkers

The effect of finerenone, compared with placebo, on the primary composite outcome, each of its components, and all-cause death according to liver biomarker levels is shown in Figure S2. The effects of finerenone on these outcomes were not modified by any of these liver biomarkers.

Change in Liver Biomarkers and Outcomes During Follow-Up

In a landmark analysis, an increase in bilirubin and ALP from baseline to 3 months after randomization was associated with a higher risk of the primary composite outcome (and other events). This was not the case for transaminases (Figure S3).

Effect of Finerenone on Liver Biomarkers During Follow-Up

Bilirubin and ALP did not change during follow-up in the placebo group. Finerenone reduced both bilirubin and ALP levels, and this reduction was observed early after

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Table 2. Outcomes According to Tertiles of Bilirubin

	Tertile 1; ≤0.4 mg/dL (n=1923)	Tertile 2; 0.5–0.6 mg/dL (n=1932)	Tertile 3; ≥0.7 mg/dL (n=2018)
Cardiovascular death and total worsening HF events			
No. of events	739	687	886
Event rate per 100 person-years (95% CI)	16.0 (14.3–17.9)	14.5 (12.9–16.3)	18.2 (16.3–20.3)
Rate ratio (95% CI)*	1.03 (0.87–1.21)	Reference	1.29 (1.10–1.51)
Rate ratio (95% CI)†	1.05 (0.90–1.24)	Reference	1.13 (0.97–1.32)
Total worsening HF events			
No. of events	593	535	698
Event rate per 100 person-years (95% CI)	12.8 (11.3–14.5)	11.3 (9.9–12.9)	14.3 (12.6–16.2)
Rate ratio (95% CI)*	1.05 (0.87–1.26)	Reference	1.31 (1.09–1.57)
Rate ratio (95% CI)†	1.07 (0.89–1.29)	Reference	1.16 (0.97–1.38)
Cardiovascular death or first worsening HF event			
No. of events, %	424 (22.0)	401 (20.8)	483 (23.9)
Event rate per 100 person-years (95% CI)	10.1 (9.2–11.1)	9.2 (8.4–10.2)	11.0 (10.0–12.0)
Hazard ratio (95% CI)*	1.04 (0.90–1.19)	Reference	1.21 (1.06–1.38)
Hazard ratio (95% CI)†	1.06 (0.92–1.22)	Reference	1.07 (0.93–1.23)
First worsening HF event			
No. of events, %	336 (17.5)	308 (15.9)	382 (18.9)
Event rate per 100 person-years (95% CI)	8.0 (7.2–8.9)	7.1 (6.3–7.9)	8.7 (7.8–9.6)
Hazard ratio (95% CI)*	1.06 (0.90–1.24)	Reference	1.25 (1.08–1.46)
Hazard ratio (95% CI)†	1.07 (0.91–1.26)	Reference	1.11 (0.95–1.29)
Cardiovascular death			
No. of events, %	146 (7.6)	153 (7.9)	189 (9.4)
Event rate per 100 person-years (95% CI)	3.2 (2.7–3.7)	3.2 (2.8–3.8)	3.9 (3.4–4.5)
Hazard ratio (95% CI)*	0.95 (0.76–1.20)	Reference	1.20 (0.97–1.49)
Hazard ratio (95% CI)†	0.96 (0.76–1.22)	Reference	1.03 (0.83–1.29)
All-cause death			
No. of events, %	332 (17.3)	302 (15.6)	349 (17.3)
Event rate per 100 person-years (95% CI)	7.1 (6.4–7.9)	6.4 (5.7–7.1)	7.1 (6.4–7.9)
Hazard ratio (95% CI)*	1.09 (0.93–1.28)	Reference	1.13 (0.97–1.32)
Hazard ratio (95% CI)†	1.13 (0.96–1.34)	Reference	0.98 (0.84–1.15)

HF indicates heart failure.

*Models were stratified by geographic region and left ventricular ejection fraction stratification and adjusted for treatment assignment.

†Models were stratified by geographic region and left ventricular ejection fraction stratification and adjusted for treatment assignment, age, sex, systolic blood pressure, heart rate, body mass index, estimated glomerular filtration rate, left ventricular ejection fraction, New York Heart Association functional class, log of N-terminal pro-B-type natriuretic peptide, log of high-sensitivity troponin T, prior heart failure hospitalization, type 2 diabetes, myocardial infarction or coronary revascularization, and atrial fibrillation/flutter.

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randomization and sustained during follow-up (ratio of geometric mean ratios at 12 months: bilirubin, 0.92 [95% CI, 0.90–0.94]; ALP, 0.96 [95% CI, 0.95–0.97]; Figure 3; Table S1).

ALT and AST decreased during follow-up in both treatment groups, with no significant difference between treatment groups (Figure 3; Table S1).

DISCUSSION

There were 5 key findings in this study. First, total bilirubin was the most frequently elevated liver biomarker in patients with HFmrEF/HFpEF participating in FINEARTS-HF. Second, bilirubin was a predictor of clinical outcomes, even after adjustment for other prognostic variables, including NT-proBNP and high-sensitivity troponin T. Third, an increase in bilirubin after randomization was associated with a higher risk of the outcomes of interest. Fourth, the benefit of finerenone was consistent across the range of bilirubin concentrations measured at baseline, including in the patients with elevations of this biomarker at high risk. Finally, compared with placebo, finerenone led to a rapid and sustained reduction in levels of bilirubin and ALP but did not affect transaminase levels.

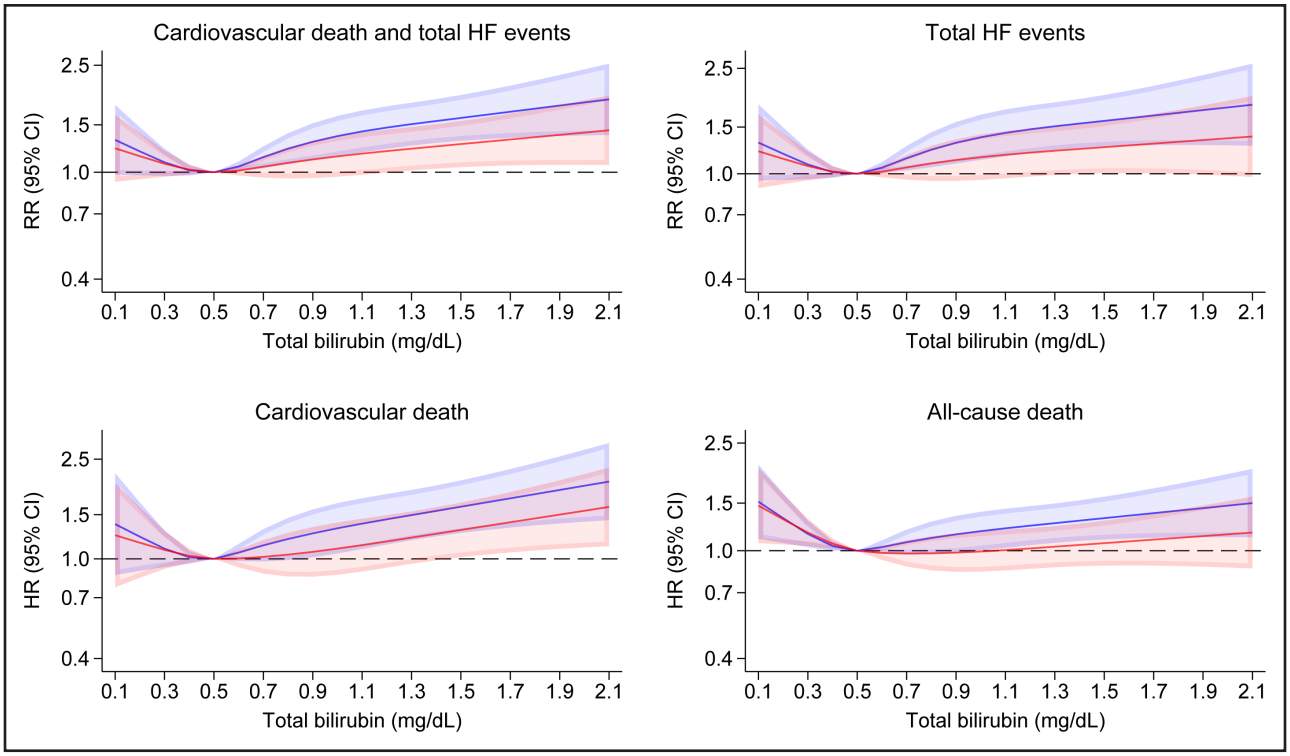

Figure 1. Outcomes according to continuous bilirubin.
The blue spline is stratified for geographic region and left ventricular ejection fraction stratification and adjusted for treatment assignment. The red spline is stratified by geographic region and left ventricular ejection fraction stratification and adjusted for treatment assignment, age, sex, systolic blood pressure, heart rate, body mass index, estimated glomerular filtration rate, left ventricular ejection fraction, New York Heart Association functional class, log of N-terminal pro-B-type natriuretic peptide, log of high-sensitivity troponin T, prior heart failure hospitalization, type 2 diabetes, myocardial infarction or coronary revascularization, and atrial fibrillation/flutter. HF indicates heart failure; HR, hazard ratio; and RR, rate ratio.

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In contrast to patients with HFrEF, relatively little is known about liver biomarkers in individuals with HFmrEF/HFpEF.[12,13] Among participants in FINEARTS-HF, the prevalence of abnormal liver tests was low, ranging from 11.9% for bilirubin to 7.8% for ALT. The proportion with an elevated bilirubin compares to 11.6% in PARADIGM-HF (Prospective Comparison of ARNI With ACEI to Determine Impact on Global Mortality and Morbidity in Heart Failure) and 14.7% in DAPA-HF (Dapagliflozin and Prevention of Adverse Outcomes in Heart Failure).[1,2]

Because transaminase levels may be more reflective of a decrease in cardiac index and liver blood flow than the elevation of central venous pressure, they might be less elevated in HFmrEF/HFpEF, compared with HFrEF.[16–24] Alternatively, the high prevalence of obesity and diabetes in patients with HFmrEF/HFpEF might result in transaminase elevations through different, nonhemodynamic mechanisms.[16–24] However, the proportion of people with an elevated level of ALT and AST was almost identical to what we found in DAPA-HF (8.3% and 7.9%, respectively) and smaller than in PARADIGM-HF (12.0% and 9.0%, respectively).[1,2]

Bilirubin seems to be an independent predictor of unfavorable clinical outcomes in patients with HFrEF, with an essentially linear increase in risk with rising bilirubin levels.[1–8] Not only that, but bilirubin remains a predictor of worse outcomes in multivariable models, including NT-proBNP and high-sensitivity troponin T, the 2 most powerful prognostic biomarkers in HFrEF.[1,2] The findings in FINEARTS-HF were slightly different, with the suggestion of a J-shaped relationship between bilirubin level and outcomes in the categorical analysis, consistent with a recent report from EMPEROR-Preserved.[12] However, when examined as a continuous variable, levels higher than the median value were associated with a greater risk of the primary outcome, each of its components, even when fully adjusted for other prognostic variables including NT-proBNP and high-sensitivity troponin T. Interestingly, an increase in bilirubin (and ALP) after randomization was associated with a higher risk of subsequent outcomes, which has not been reported in prior studies. This was not the case for increases in transaminases.

It is most likely that bilirubin reflects central venous/right atrial pressure in patients with HF, and we have postulated that it provides different information about central hemodynamics than NT-proBNP, which may be more reflective of left- than right-sided pressures.[9–11] Because bilirubin is associated with right-sided filling pressures, we anticipated that finerenone would reduce

Table 3. Effects of Finerenone Compared With Placebo on Outcomes According to Tertiles of Bilirubin

	Tertile 1; ≤0.4 mg/dL (n=1923)		Tertile 2; 0.5–0.6 mg/dL (n=1932)		Tertile 3; ≥0.7 mg/dL (n=2018)		
	Finerenone; n=957	Placebo; n=966	Finerenone; n=970	Placebo; n=962	Finerenone; n=1009	Placebo; n=1009	$P_{interaction}$
Cardiovascular death and total worsening HF events							0.43
No. of events	355	384	305	382	398	488	
Event rate per 100 person-years (95% CI)	15.5 (13.1–18.3)	16.5 (14.1–19.2)	12.9 (11.0–15.2)	16.1 (13.7–19.0)	16.2 (13.6–19.2)	20.2 (17.5–23.3)	
Rate ratio (95% CI)*	0.94 (0.75–1.17)		0.83 (0.66–1.05)		0.77 (0.62–0.97)		
Total worsening HF events							0.42
No. of events	284	309	230	305	311	387	
Event rate per 100 person-years (95% CI)	12.4 (10.2–14.9)	13.2 (11.1–15.7)	9.7 (8.0–11.8)	12.9 (10.7–15.5)	12.6 (10.4–15.3)	16.0 (13.6–18.9)	
Rate ratio (95% CI)*	0.94 (0.73–1.21)		0.79 (0.60–1.03)		0.76 (0.59–0.97)		
Cardiovascular death or worsening HF event							0.06
No. of events, %	199 (20.8)	225 (23.3)	195 (20.1)	206 (21.4)	212 (21.0)	271 (26.9)	
Event rate per 100 person-years (95% CI)	9.5 (8.3–10.9)	10.7 (9.3–12.1)	8.9 (7.7–10.3)	9.5 (8.3–10.9)	9.3 (8.2–10.7)	12.7 (11.3–14.3)	
Hazard ratio (95% CI)*	0.89 (0.73–1.08)		0.96 (0.79–1.17)		0.71 (0.60–0.86)		
First worsening HF event							0.42
No. of events, %	155 (16.2)	181 (18.7)	140 (14.4)	168 (17.5)	171 (16.9)	211 (20.9)	
Event rate per 100 person-years (95% CI)	7.4 (6.3–8.6)	8.6 (7.4–9.9)	6.4 (5.4–7.6)	7.8 (6.7–9.0)	7.5 (6.5–8.8)	9.9 (8.6–11.3)	
Hazard ratio (95% CI)*	0.87 (0.70–1.07)		0.86 (0.68–1.07)		0.73 (0.60–0.90)		
Cardiovascular death							0.76
No. of events, %	71 (7.4)	75 (7.8)	75 (7.7)	78 (8.1)	88 (8.7)	101 (10.0)	
Event rate per 100 person-years (95% CI)	3.1 (2.4–3.9)	3.2 (2.6–4.0)	3.2 (2.5–4.0)	3.3 (2.6–4.1)	3.6 (2.9–4.4)	4.2 (3.4–5.1)	
Hazard ratio (95% CI)*	0.95 (0.68–1.31)		0.97 (0.71–1.34)		0.85 (0.64–1.13)		
All-cause death							0.62
No. of events, %	158 (16.5)	174 (18.0)	151 (15.6)	151 (15.7)	164 (16.3)	185 (18.3)	
Event rate per 100 person-years (95% CI)	6.8 (5.8–8.0)	7.4 (6.4–8.6)	6.4 (5.4–7.5)	6.4 (5.4–7.5)	6.6 (5.7–7.7)	7.6 (6.6–8.8)	
Hazard ratio (95% CI)*	0.92 (0.74–1.14)		1.00 (0.80–1.26)		0.85 (0.69–1.05)		
Improvement in NYHA class from baseline to 12 mo							0.70
N (%)	158 (16.5)	159 (16.5)	175 (18.0)	180 (18.7)	211 (20.9)	197 (19.5)	
Odds ratio (95% CI)†	1.01 (0.79–1.28)		0.95 (0.76–1.20)		1.10 (0.88–1.37)		
Change in KCCQ-TSS from baseline to 12 mo							0.11
Mean change (SD)	8.08 (20.81)	7.99 (21.19)	10.60 (21.82)	7.99 (21.11)	5.44 (24.65)	7.78 (24.02)	
Difference in mean (95% CI)‡	1.13 (-0.62–2.89)		0.69 (-0.92–2.30)		3.15 (1.52–4.77)		

HF indicates heart failure; KCCQ-TSS, Kansas City Cardiomyopathy Questionnaire total symptom score; and NYHA, New York Heart Association.
*Models were stratified for geographic region and left ventricular ejection fraction stratification.
†Models were adjusted for geographic region and left ventricular ejection fraction stratification.
‡Models were adjusted for baseline value, geographic region, and left ventricular ejection fraction stratification.

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bilirubin by alleviating hepatic congestion through its direct diuretic action or by augmentation of the effect of concomitant loop or thiazide diuretic therapy. Finerenone did decrease bilirubin levels, significantly, by 8%. It is of interest to compare this with the effects of sacubitril/valsartan, which, compared with enalapril, also reduced bilirubin significantly in PARADIGM-HF by a small amount (2.4%), and SGLT2 inhibitors, which did not reduce bilirubin in HF.[1,2,12]

Although the reduction in bilirubin levels with finerenone may reflect a decongestive effect, other mechanisms should also be considered. As mentioned in the

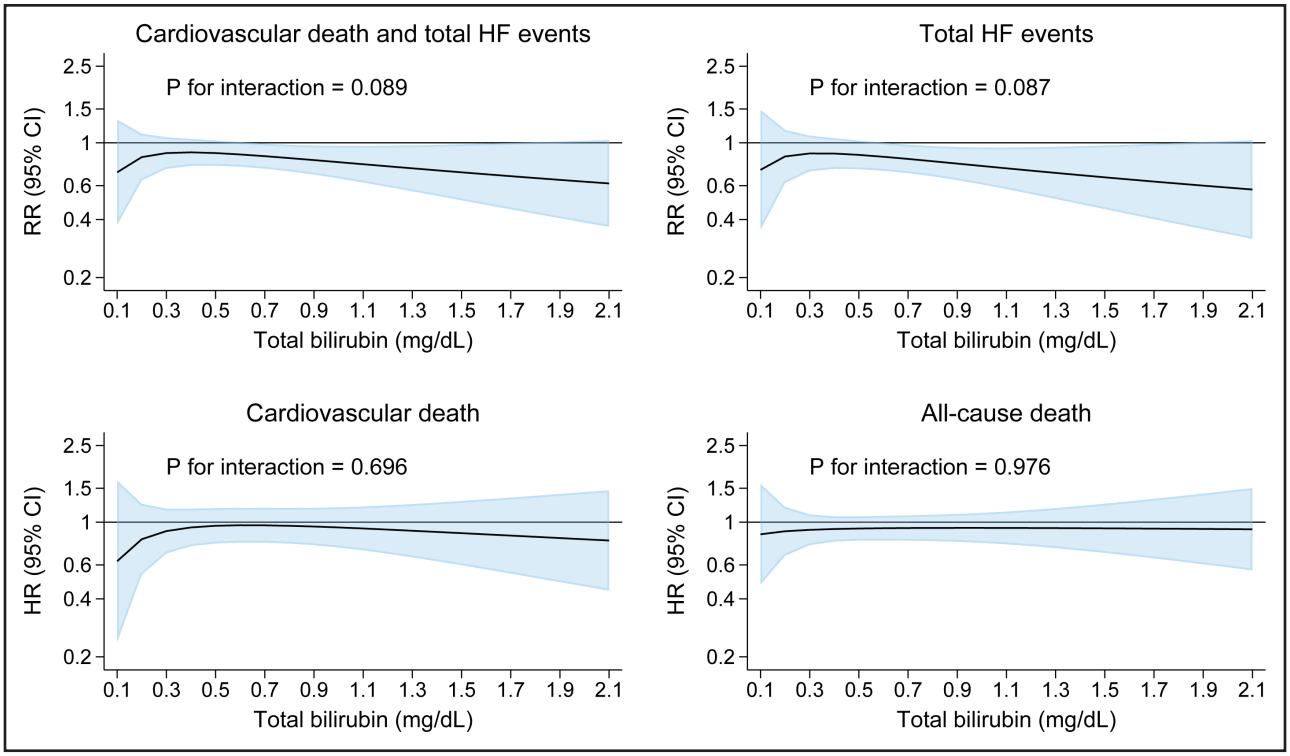

Figure 2. Effects of finerenone compared with placebo on outcomes according to continuous bilirubin.
Models were stratified for geographic region and left ventricular ejection fraction stratification. The *P* value reflects the interaction between the bilirubin concentration at baseline and the treatment effect. A *P*>0.05 indicates a nonsignificant interaction, that is, that bilirubin at baseline does not modify the effect of finerenone vs placebo. HF indicates heart failure; HR, hazard ratio; and RR, rate ratio.

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Introduction, activation of the mineralocorticoid receptor, which is also expressed in hepatic tissue, may promote hepatic steatosis, inflammation, and fibrosis, and antagonizing this receptor has been shown to attenuate these pathological processes by reducing oxidative stress, hepatic fibrosis, and insulin resistance.[16–21] In addition, preclinical studies have suggested that spironolactone may alter bilirubin conjugation, potentially influencing bilirubin levels independently of hemodynamic changes.[30–33] These data raise the possibility that finerenone may exert direct hepatic effects in addition to a potential decongestive effect.

In contrast to bilirubin, elevated transaminase levels were not associated with worse outcomes in FINEARTS-HF, consistent with recent reports in both HFrEF and HFmrEF/HFpEF.[1,2,12,13] Although we hypothesized that the anti-inflammatory and antifibrotic actions of finerenone might reduce transaminase levels, we did not see this in FINEARTS-HF. Another study with finerenone in patients with chronic kidney disease and diabetes recently reported similar findings.[34] This benefit was also anticipated with SGLT2 inhibitors but not observed with these drugs either.[2,12] By contrast, sacubitril/valsartan did reduce AST and ALT in PARADIGM-HF, suggesting either a specific effect of neprilysin inhibition on transaminase activity or, more likely, a greater effect of sacubitril/valsartan on cardiac index in HFrEF, compared with finerenone or an SGLT2 inhibitor in HFmrEF/HFpEF.[1]

Interestingly, ALP was also independently predictive of outcomes in FINEARTS-HF, as has been observed in HFrEF.[1,2] Although ALP is a less specific hepatic biomarker (as it may also originate from bone and other tissues as well), the similar time course of the reduction in ALP and bilirubin (in contrast to the lack of change in transaminases) suggests that ALP also reflects the effect of right-sided filling pressures on the liver and maybe other tissues.

Remarkably, there are no data published about the effect of either spironolactone or eplerenone on liver tests in HF and, therefore, no comparison with finerenone is possible.

Baseline bilirubin level did not modify the effect of finerenone, as was also observed with empagliflozin in HFmrEF/HFpEF and both sacubitril/valsartan and dapagliflozin in HFrEF.[1,2,12] Because of the high risk related to elevated bilirubin levels, the absolute benefits of all therapies in patients with raised bilirubin levels are large.

Study Limitations

This was not a prespecified analysis. Because the patients studied were in a clinical trial, the specific enrollment criteria limit the generalizability of our

Table 4. Effects of Finerenone Compared With Placebo on Laboratory Measures and Systolic Blood Pressure According to Tertiles of Bilirubin

	Tertile 1; ≤0.4 mg/dL (n=1917)		Tertile 2; 0.5–0.6 mg/dL (n=1929)		Tertile 3; ≥0.7 mg/dL (n=2013)		
	Finerenone; n=955	Placebo; n=962	Finerenone; n=968	Placebo; n=961	Finerenone; n=1004	Placebo; n=1009	$P_{interaction}$
Creatinine ≥2.5 mg/dL							0.75
No. of events, %	60/920 (6.5)	39/923 (4.2)	39/933 (4.2)	25/934 (2.7)	39/978 (4.0)	19/973 (2.0)	
Odds ratio (95% CI)*	1.63 (1.07–2.47)		1.59 (0.95–2.66)		2.06 (1.18–3.61)		
Creatinine ≥3 mg/dL							0.93
No. of events, %	28/920 (3.0)	16/923 (1.7)	14/933 (1.5)	9/934 (1.0)	12/978 (1.2)	7/973 (0.7)	
Odds ratio (95% CI)*	1.82 (0.98–3.40)		1.51 (0.65–3.53)		1.77 (0.69–4.54)		
Potassium >5.5 mmol/L							0.44
No. of events, %	160/919 (17.4)	74/924 (8.0)	114/933 (12.2)	63/935 (6.7)	128/980 (13.1)	55/972 (5.7)	
Odds ratio (95% CI)*	2.40 (1.79–3.22)		1.97 (1.42–2.73)		2.65 (1.90–3.71)		
Potassium >6 mmol/L							0.48
No. of events, %	36/919 (3.9)	12/924 (1.3)	26/933 (2.8)	16/935 (1.7)	23/980 (2.3)	10/972 (1.0)	
Odds ratio (95% CI)*	3.00 (1.55–5.82)		1.70 (0.90–3.21)		2.43 (1.14–5.16)		
Potassium <3.5 mmol/L							0.24
No. of events, %	41/919 (4.5)	72/924 (7.8)	37/933 (4.0)	88/935 (9.4)	46/980 (4.7)	115/972 (11.8)	
Odds ratio (95% CI)*	0.55 (0.37–0.81)		0.41 (0.27–0.60)		0.35 (0.24–0.50)		
Systolic blood pressure <100 mmHg							0.04
No. of events, %	137/926 (14.8)	118/930 (12.7)	169/936 (18.1)	115/938 (12.3)	220/984 (22.4)	123/977 (12.6)	
Odds ratio (95% CI)*	1.24 (0.94–1.63)		1.76 (1.34–2.31)		2.02 (1.56–2.61)		

A total of 14 randomized patients were excluded from the safety analysis, as these were performed in patients who had undergone randomization and received at least one dose of finerenone or placebo.

*Models were adjusted for geographic region and left ventricular ejection fraction stratification.

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findings. However, the only hepatic exclusion criterion was patients with advanced liver disease (Child-Pugh grade C). Other measures reflecting hepatic function, including albumin and international normalized ratio, were not made in FINEARTS-HF, and other relevant enzymes, including gamma-glutamyl transferase, were

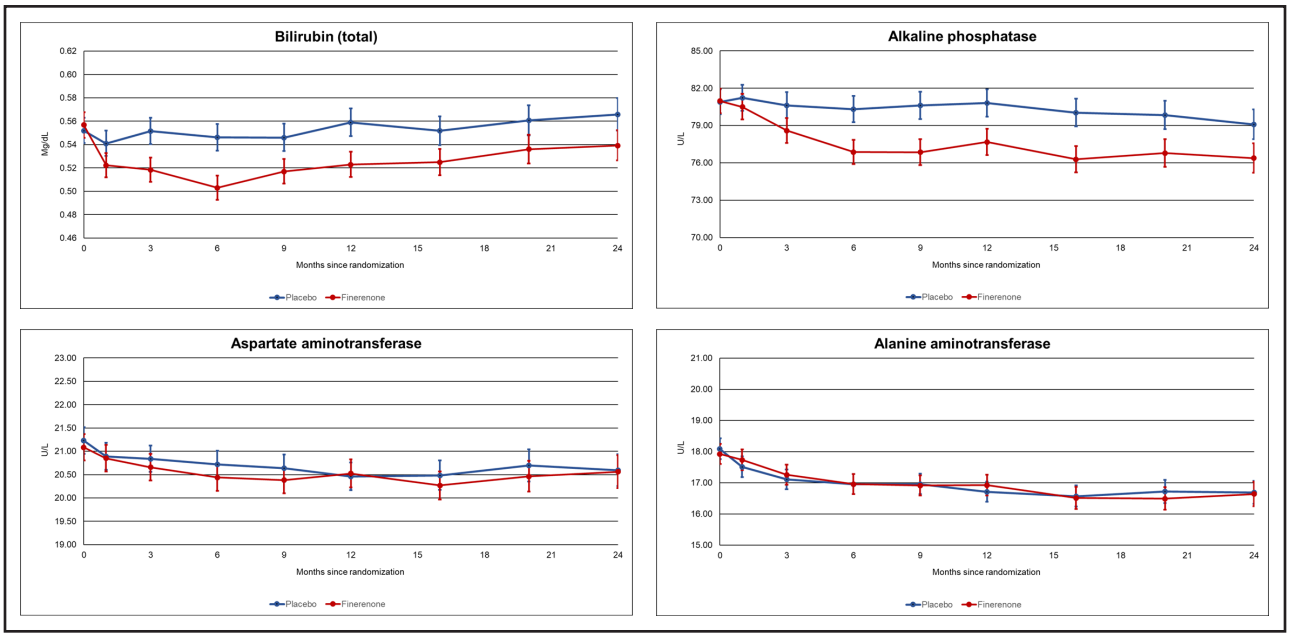

Figure 3. Change in markers of liver function with finerenone compared with placebo from baseline to 24 months. Dots represent geometric means, and error bars represent 95% CIs.

not measured. We did not collect information on right-sided filling pressures, and echocardiographic data on right heart structure and function were not available. Finally, in the analyses of the association between baseline bilirubin levels (and other liver biomarkers) and outcomes, the risk of residual confounding cannot be excluded despite comprehensive adjustment for potential confounders.

Conclusions

In patients with HFmrEF/HFpEF, baseline bilirubin concentration was an independent predictor of worse outcomes, but it did not modify the benefits of finerenone on morbidity and mortality. Finerenone led to a rapid and sustained reduction in levels of bilirubin and ALP.

ARTICLE INFORMATION

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Disclosures

Dr Butt reports advisory board honoraria from AstraZeneca and Bayer; consultant honoraria from Novartis and AstraZeneca; and travel grants from AstraZeneca. Dr Jhund reports speakers' fees from AstraZeneca, Novartis, Alkem Metabolics, ProAdWise Communications, Sun Pharmaceuticals; advisory board fees from AstraZeneca, Boehringer Ingelheim, Novartis; and research funding from AstraZeneca, Boehringer Ingelheim, Analog Devices Inc, Roche Diagnostics. Dr Jhund's employer the University of Glasgow has been remunerated for clinical trial work from AstraZeneca, Bayer AG, Novartis, and Novo Nordisk; and Dr Jhund is the Director of Global Clinical Trial Partners Ltd. Dr Claggett has received personal consulting fees from Alnylam, Bristol Myers Squibb (BMS), Cardior, Cardurion, Corvia, CVRx, Eli Lilly, Intellia, and Rocket and has served on a data safety monitoring board for Novo Nordisk. 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Dr Pitt is a consultant for Bayer, Astra Zeneca, Boehringer Ingelheim, Lexicon, Bristol Meyers Squibb, KBP Biosciences (stock/stock options), Sarfez Pharmaceuticals (stock/stock options), Pharmaceuticals (stock/stock options), SQinnovations (stock/stock options), G3 Pharmaceuticals, Sea Star medical (stock/stock options), Vifor (stock/stock options), Prointel (stock/stock options), and Brainstorm Medical (stock/stock options), US Patent 9931412-site specific delivery of eplerenone to the myocardium, US Patent pending 63/045,783 Histone modulating agents for the prevention and treatment of organ failure. 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Supplemental Material

Table S1
Figures S1–S3

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